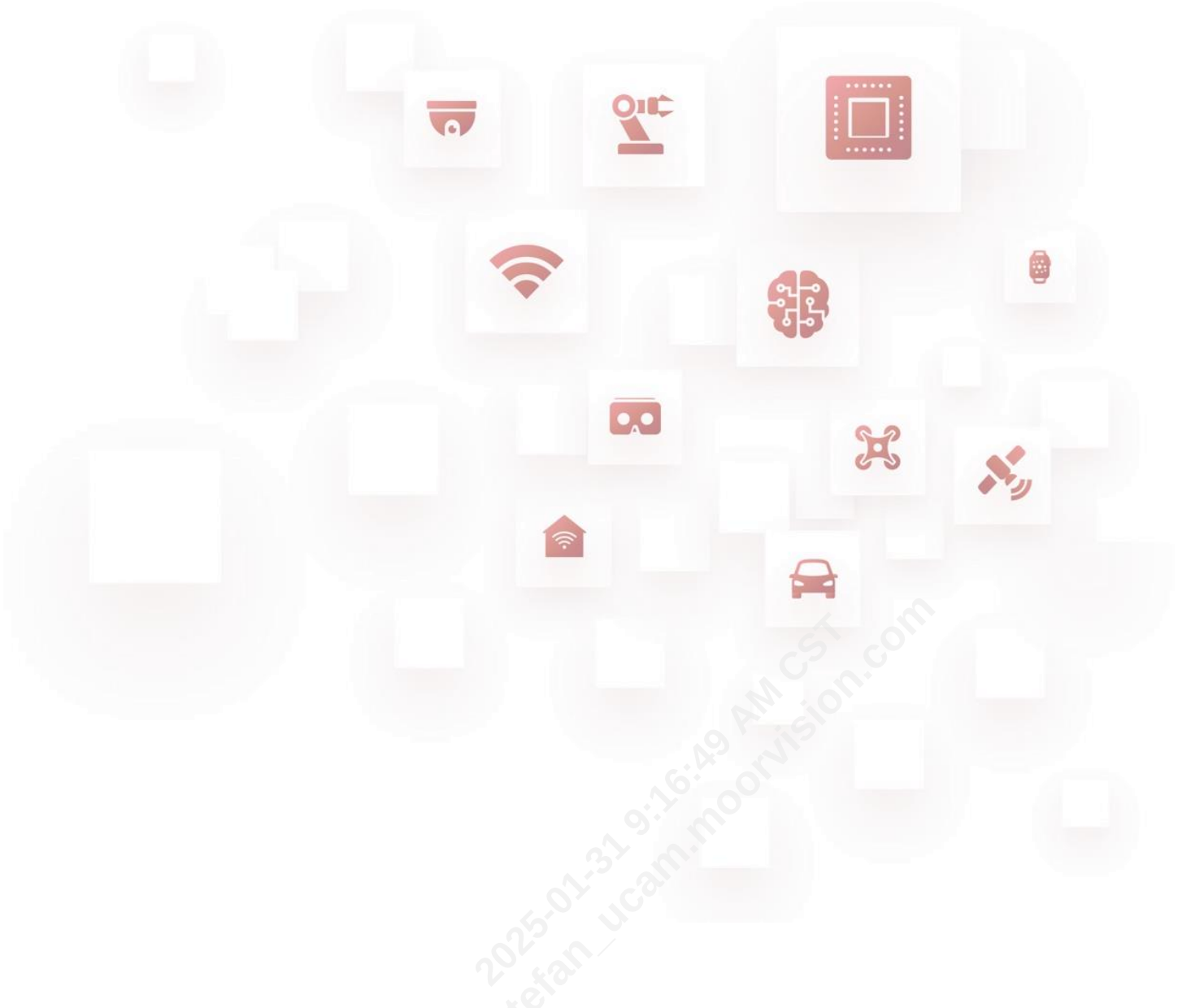
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Thundercomm TurboX C410/C610 Open Kit Linux Android

Software Release Notes

Rev. V1.0
Sep 20, 2022

DN: tc-gla-2410
SDK Version: LA2.0.CS.r002001

Revision History

Revision	Date	Description
1.0	Sep 20, 2022	Initial release.

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About This Document

- Illustrations in this documentation might look different from your product.
- Depending on the model, some optional accessories, features, and software programs might not be available on your device.
- Depending on the version of operating systems and programs, some user interface instructions might not be applicable to your device.
- Documentation content is subject to change without notice. Thundercomm makes constant improvements on the documentation of the products, including this guidebook.
- Function declarations, function names, type declarations, attributes, and code samples appear in a different format, for example, `cp armcc armcpp`.
- Code variables appear in angle brackets, for example, `<number>`.
- Button, tool, and key names appear in bold font, for example, click **Save** or press **Enter**.
- Commands to be entered appear in a different font; on the host computer use $ as shell prompt, while on the target device use # as shell prompt, for example,

```
$ adb devices
# logcat
```

- Part of the code that does not contain instructions appear in a different format, for example,

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
```

- Folders and files are formatted in italic, for example, *turbox_flash_flat.sh*.

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Chapter 1. Introduction

This document provides information on how to obtain, build, and program QCS410/610 LA (Linux Android) SDK (Software Development Kit) into the TurboX C410/C610 Open Kit.

➲ **NOTE:** "TurboX" referred to herein is the English text of our registered trademark TURBOX.

The Linux Android Software Development Kit supports the Android operating system on the Thundercomm TurboX. The goal of the LA SDK is to provide a simple and flexible development platform to rapidly build IoT products with the QTI chipsets. The LA SDK provides the build framework, software components, tools, and sample applications to evaluate the platform and build software images for the target product.

➲ **NOTE:** The LA Software can run on the 4GB DDR TurboX C610 platform with best performance.

Table 1-1. TurboX C410/C610 platform specifications

SOC	SKUx		Standard SKU[1)]	OS	Thundercomm Product No. (TPN)	Model
QCS610	SKU1	2+16GB	Yes	LA[2)]	C610-D21-369PWXX C610-D21-ACXPW00 (LA Specified)	TurboX C610
	SKU2	2+64GB	No	LA[2)]	C610-D26-369PWXX C610-D26-ACXPW00 (LA Specified)	TurboX C610
	SKU3	4+64GB	Yes	LA	C610-D46-369PWXX C610-D46-ACXPW00 (LA Specified)	TurboX C610
QCS410	SKU4	2+16GB	Yes	LA[2)]	C410-D21-369PWXX C610-D46-ACXPW00 (LA Specified)	TurboX C410
QCS410	SKU5	2+64GB	No	LA[2)]	C410-D26-369PWXX C410-D26-ACXPW00 (LA Specified)	TurboX C410
QCS410	SKU7	4+64GB	No	LA	C410-D46-369PWXX C410-D46-ACXPW00(LA Specified)	TurboX C410

➲ **NOTE:**

[1)] indicates standard SKU series. For standard SKU series, retail stock is available; as to non-standard SKU series, in case of any customer requirement, specific orders need to be placed accordingly.

[2)] indicates that when the memory is less than 4GB, Android 12 OS runs out of memory. In this case, it is necessary to tailor the system as required to stabilize the system.

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Chapter 2. Development Environment Setup

Refer to *TurboX SOM SDK Manager User Guide* to set up the development environment.

➲ **NOTE:** Make sure that you use the sdkmanager built based on Ubuntu 18.04. When you start the sdkmanager, the following window will appear.

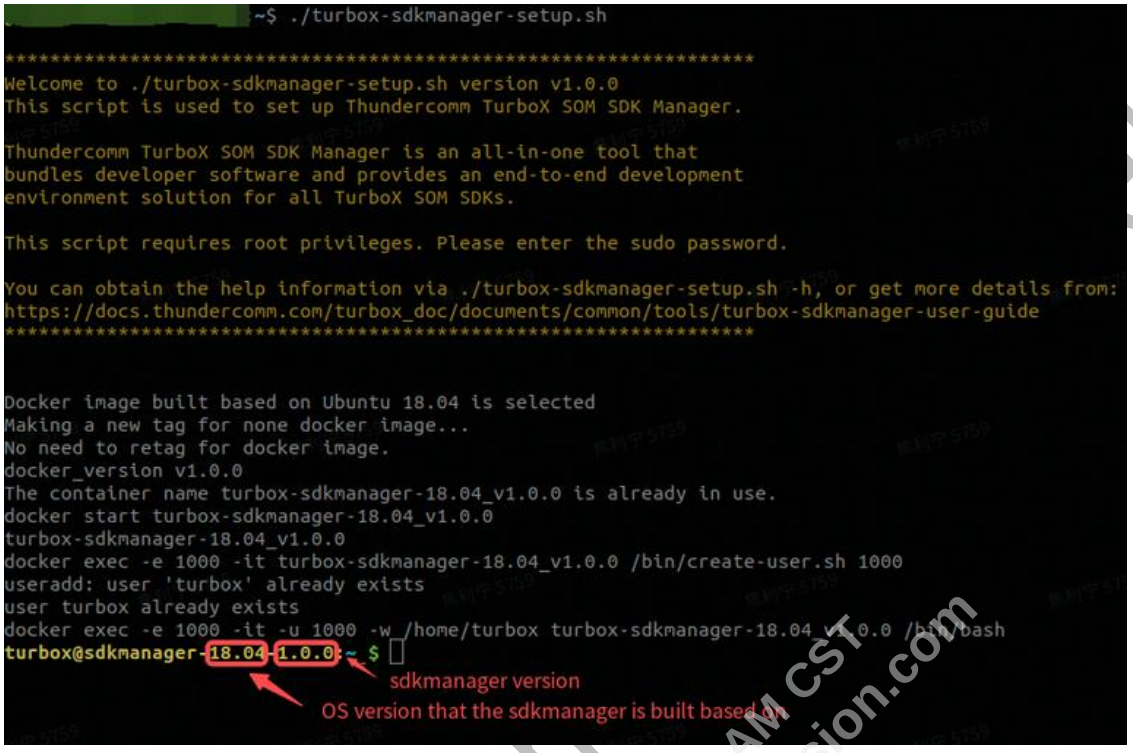


Figure 2-1.sdkmanager Window

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Chapter 3. Get Started with SDK

3.1. SDK information

Table 3-1 presents the LA SDK information, including the SDK version.

Table 3-1. LA SDK information

SDK Version	Baseline	Baseline Distribution	OS Version
LA2.0.CS.r002001	QCS610.LA.2.0	CS[1] 0.0.008.1	Android 12.00

3.2. SDK block diagram

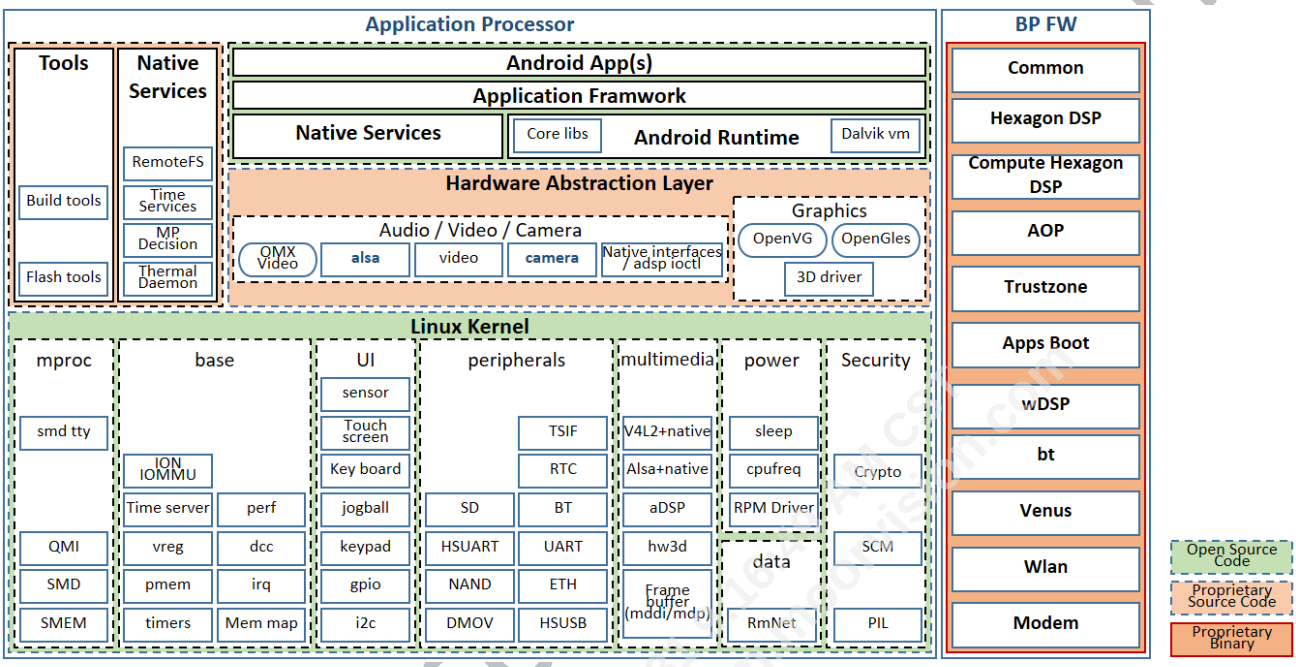


Figure 3-1. Block Diagram

3.3. SDK source tree structure

```
SDK.Turbox_C610.LA.2.0/
├── about.html -> common/about.html            //SDK version info(Product,
Distribution,Components, …)
├── contents.xml -> common/contents.xml        //image contents file
├── common/config/emmc/partition.xml           //eMMC partition file
├── LINUX
│   └── android
│       ├── bionic
│       ├── bootable/bootloader/edk2           //bootloader sources
│       ├── bootable/recovery                  //recovery sources
│       ├── build              //basic config files for development kit, such as files for
system build rules and generis
│       ├── dalvik
│       ├── developers                 //developer directory
│       ├── development                //infrastructure-related files for Android application
development
│       └── device                          //device related config
│           └── qcom/sm6150                 //QCS610 device-related config directory
│       ├── external                        //files related to open-source development kit
│       ├── frameworks                      //source code under Android frameworks
│       ├── hardware/qcom                   //code for QCS610 hardware abstraction layer
 (bootctrl, Display, gps, media, vr, …)
```

```
│       ├── kernel
│       │   └── msm-5.4                                  //kernel sources
│       │       ├──arch/arm64/boot/dts/vendor/qcom       //DT source files
│       │       └──arch/arm64/configs/vendor             //kernel config files
│       ├── libcore                                      //core library files
│       ├── libnativehelper          //dynamic library, which is the basis to JNI library
│       ├── Makefile                                  //global Makefile to define build rules
│       ├── packages                                  //application packages
│       ├── pdk                                       //plug development kit
│       ├── platform_testing                          //platform testing files
│       ├── prebuilts                //prebuild resources under x86 and arm framework
│       ├── sdk                                       //sdk and simulator
│       ├── system                   //underlying file system, application, and component
│       ├── test                                      //Android Vendor testing framework
│       ├── toolchain                                 //tool chain files
│       ├── tools                                     //tools files
│       └── vendor                                    //QCOM private repository
│           └── qcom
│               ├── opensource
│               └── proprietary
├── turbox_build.sh -> turbox/tools/build_script/common/turbox_build.sh       //SDK build
file.
├── turbox_flash.sh -> turbox/tools/build_script/device/turbox_flash.sh       //Fastboot
flash tool
├── turbox
│   └── tools
│       ├── build_script          //SDK build script(android build, zip, …)
│       └── flash_tools           //Flash tools files(qdl, …)
├── adsp_proc             //aDSP image is released as a binary and no build compilation is
needed
├── aop_proc              //AOP image is released as a binary and no build compilation is
needed
├── boot_images           //BOOT image is released as a binary and no build compilation is
needed
├── btfm_proc             //BTFM image is released as a binary and no build compilation is
needed
├── btfm_proc_CMC         //BTFM_CMC image is released as a binary and no build
compilation is needed
├── cdsp_proc             //cDSP image is released as a binary and no build compilation is
needed
├── modem_proc            //MODEM image is released as a binary and no build compilation
is needed
├── trustzone_images      //TZ image is released as a binary and no build compilation is
needed
├── venus_proc            //VIDEO image is released as a binary and no build compilation
is needed
├── wdsp_proc             //wDSP image is released as a binary and no build compilation is
needed.
└── wlan_proc             //WLAN image is released as a binary and no build compilation is
needed.
```

3.4. Download SDK

Follow the steps below to activate an account and download the SDK code.

3.4.1. Configure git global variable

Configure user name and email.

```
turbox@sdkmanager-18.04-1.2.0:~ $ git config --global user.email "<User Email>"
turbox@sdkmanager-18.04-1.2.0:~ $ git config --global user.name "<User Name>"
```

➲ **NOTE:**

- Please use the email and user name you have registered at *Thundercomm website* to replace `<User Email>` and `<User Name>` in the above commands.
- Do not enter the angle brackets "`<>`" which are used for indicating code variables.

3.4.2. Account activation

Add the account/password information to *$HOME/.netrc* by running the following command.

```
turbox@sdkmanager-18.04-1.2.0:~ $ echo "machine partner.thundercomm.com login <User Email>
password <password>" > ~/.netrc
```

➲ **NOTE:** Please use the email/password you have registered at *www.thundercomm.com* website to replace `<User Email>` and `<password>` in the above command.

3.4.3. Download the SDK code

Follow the steps below to download the SDK code.

Step 1.Initialize the repository.

```
turbox@sdkmanager-18.04-1.2.0:~ $ repo init -u
https://partner.thundercomm.com/sdkgit/a/manifest -b SDK.Turbox-QCS610.LA.2.0 -m
turbox-c610-la2.0-dev.release.CS.r002001.xml --repo-
url=https://partner.thundercomm.com/sdkgit/a/tools/repo --repo-branch=stable --no-
repo-verify
```

Step 2.Synchronize the source code.

```
turbox@sdkmanager-18.04-1.2.0:~ $ repo sync -c -j4 --no-tags
```

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3.4.4. SDK version update

If you have downloaded the old-version SDK, there are two ways to download the new SDK fast.

Option 1. Upgrade the local old folder to a new release

Navigate to the old local code folder, you can run the repo init command of new release as specified in *3.4.3. Download the SDK code*, and then run `repo sync -j4 -c --no-tags` to receive the new code.

Option 2. Download the new release to a new folder(recommended)

Create a new folder, and run `repo init` with `-reference` value.

For example:

```
turbox@sdkmanager-18.04-1.2.0:~ $ repo init -u
https://partner.thundercomm.com/sdkgit/a/manifest -b SDK.Turbox_C610.LA2.0 -m
turbox-c610-la2.0-dev.release.CS.r002001.xml --repo-
url=https://partner.thundercomm.com/sdkgit/a/tools/repo --repo-branch=stable --
no-repo-verify --reference=<old_sdk_code_path>
$ repo sync -j4 -c --no-tags
```

➲ **NOTE:** In `--reference=<old_sdk_code_path>`, the `old_sdk_code_path` is the absolute path of your old sdk folder.

3.5. Build images

To compile code, run the following script file in the root directory of source code for building userdebug version.

```
turbox@sdkmanager-18.04-1.2.0:~ $ ./turbox_build.sh -a -l -v userdebug
```

➲ **NOTE:** To build user version, replace `-v userdebug` with `-v user` in the above command. For detailed usage method, input `-h` to view the help information.

3.6. Pack ROM package

Follow the steps below to pack ROM package for userdebug version.

➲ **NOTE:** If you desire to pack ROM package for user version, please replace `-v userdebug` with `-v user` in the following commands.

Step 1. Run the following command to update common info for userdebug version.

```
turbox@sdkmanager-18.04-1.2.0:~ $ ./turbox_build.sh -u -v userdebug
```

➲ **NOTE**: The command executes successfully with the following log displaying on shell.

```
- update_common_info.py:======  UPDATE COMMON INFO COMPLETE=======
```

Step 2. Run the following command to pack the ROM package. You can find the ROM package at *turbox/output*.

- Pack **Fastboot** image

```
turbox@sdkmanager-18.04-1.2.0:~ $ ./turbox_build.sh --zip_fastboot_build -l -v
userdebug
```

- Pack **Flat** image

```
turbox@sdkmanager-18.04-1.2.0:~ $ ./turbox_build.sh --zip_flat_build -l -v
userdebug
```

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➲ **NOTE:** Customers can also get ROM package provided by Thundercomm from:

- **For userdebug version:**
  - *FastbootBuild_TurboX_C610_xx.xx_LA2.0.R.userdebug.CS.r002001.zip*
    MD5: 074373614b035d47719cf5db1375cb11
  - *FlatBuild_TurboX_C610_xx.xx_LA2.0.R.userdebug.CS.r002001.zip*
    MD5: 1c2e1d0128228d3e334aff129e5fa708
- **For user version:**
  - FastbootBuild_TurboX_C610_xx.xx_LA2.0.R.user.CS.r002001.zip
    MD5: be9269f72abe93b151ec2940e7c7369c
  - FlatBuild_TurboX_C610_xx.xx_LA2.0.R.user.CS.r002001.zip
    MD5: 7d13850c76efce0993b270595b869114

3.7. Flash images to board

➲ **NOTE**: If the device fails to boot up after image flash with **Fastboot**, try to flash the **FlatBuild** image again with *turbox_flash_flat.sh or TurboX Easy Flash*.

3.7.1. Flash images on Windows

You can flash images on Windows by using TurboX Easy Flash, which is a tool created by Thundercomm to help SOM developers easily flash device image, backup and restore data, etc.

For tool acquisition and operation procedures, refer to *TurboX Easy Flash User Guide*.

3.7.2. Flash images on Linux

- **Flash images on Linux with Fastboot**

Step 1. Enter the **Fastboot** mode.

Option 1. Enter the following command when the device in **ADB** mode.

```
$ adb root
$ adb reboot bootloader
```

Option 2. Power off the device, hold the **Volume down** button and click the **Power on** button, then release the **Volume down** button about 5 seconds later.

➲ **NOTE**:

To check if the device has entered the **Fastboot** mode, run the following command:

```
$ sudo fastboot devices
```

Successful with the following output appearing on the shell.

```
21f10525   fastboot
```

Step 2. Flash images.

Execute the commands below to flash images to the board. Replace `<folder name of image package>` with the actual image package name of your product.

```
$ cd <folder name of image package>
$ sudo ./turbox_flash.sh --all -r -f -v <ROM type>
```

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➲ **NOTE:**

- Please replace **<ROM type>** with **<debug>** (for userdebug version) or **<user>** (for user version) as required. Do not enter the angle brackets "**<>**" which are used for indicating code variables.
- Enter **./turbox_flash.sh -h** to acquire detailed usage.

```
$ ./turbox_flash.sh -h
Usage:
  bash ./turbox_flash.sh [options] [options]
  Images except gpt,userdata,persist will be flashed if no options input

Description:
  flash images with fasboot

Options:
  -h, --help   display this help message
  -v, --variant <debug/perf/user>    set the build variant.(default: debug)
  -f, --force  forcibly flash gpt,persist and userdata without confirmation.
               User data and calibration data(BT/WIFI MAC address, etc..) will
be ereased.
  -r, --reboot      reboot the device automatically after flashing completes.
  -b, --nonhlos     flash all bp images(xbl,aop,ftfm,tz,dsp,modem,pmic).
  -a, --hlos        flash all ap
images(aboot,bootimage,recovery,system,vendor,cache,mdtp,userdata,persist).
  --all        flash all images

Example:
  ./turbox_flash.sh
      Flash the images except gpt,userdata,persist.

  ./turbox_flash.sh --force
      Flash all images.

  ./turbox_flash.sh -hlos
Flash all ap images.
```

- **Flash images on sdkmanager with *turbox_flash_flat.sh* tool**

➲ **NOTE:** Please update the sdkmanager version to v1.2.0 which has built-in *turbox_flash_flat.sh*.

Step 1. Stop the ModemManager on your host PC and close the application that occupies the ttyUSB0 port, such as **Minicom**.

```
$ sudo systemctl stop ModemManager
```

Step 2. Prepare the FlatBuild images.

Check if the images are ready and exists in sdkmanager, then enter the image folder. Replace with the actual image package name of your product.

```
turbox@sdkmanager-18.04-1.2.0:~/workspace $ cd <folder name of image package>
```

Step 3. Enter 9008 Mode.

1) Power off the the device by disconnecting the power cable and USB cable.

2) Connect the power cable.

3) Hold the **Force USB Boot** button.

4) Press the **Power on** button and wait for about 3 seconds.

5) Release the **Force USB Boot** button and **Power on** button.

6) Connect the Type-C USB cable to an Ubuntu computer.

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To check if your device has entered the **9008 Mode**, run the following command:

```
turbox@sdkmanager-18.04-1.2.0:~ $ lsusb
……
Bus 001 Device 042: ID 05c6:9008 Qualcomm, Inc. Gobi Wireless Modem (QDL mode)
…… .
```

Step 4. Start to flash the FlatBuild images.

Run the following command under the image folder.

```
turbox@sdkmanager-18.04-1.2.0:~/workspace/<folder name of image package> $ turbox-flash-flat.sh
```

If the flashing process is successful, the following log will be printed.

```
11:20:09: INFO:       _                (done)
11:20:09: INFO:      | |
11:20:09: INFO:    __| | ___   _ __    ___
11:20:09: INFO:   / _` |/ _ \| '_ \ / _ \
11:20:09: INFO: | (_| | (_) | | | |  __/
11:20:09: INFO:  \__,_|\___/|_| |_|\___|
11:20:09: INFO: {All Finished Successfully}

11:20:09: INFO: Overall to target 30.119 seconds (85.78 MBps)
11:20:09: INFO: {percent files transferred 100.00%}

Writing log to '/home/turbox/workspace/FlatBuild_Turbox-
C610_xx.xx_la2.0.D.userdebug.20220912.2252/port_trace.txt', might take a minute
```

The device board will automatically reboot.

Connect the USB 3.1 Gen 1 Type-C cable and check the adb device with the following commands.

```
$ adb devices
List of devices attached
ac55dc38        device
```

➲ **NOTE:** For more information, please go to: https://docs.thundercomm.com/turbox_doc/documents/common/tools/turbox_flash_flat-user-guide.

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Chapter 4. Features

This chapter introduces the software features supported by the DK.

4.1. Presented features

This section describes the key software features that have been included in earlier versions.

- Kernel
  - Linux Kernel 5.4
  - 64-bit only
- OS
  - Android 12
  - F2FS + FBE
  - Dynamic partition
  - Fastboot support
  - Boot image header version 1
  - Camera 2.0 (HAL 3.4)
  - Keymaster 4.0
  - Vulkan 1.1
  - OTA Android S to S
  - Incremental OTA
- CV
  - FastCV
- Security
  - Secure boot
  - Secure video encoding
  - Secure camera
  - Secure DSP
- Connectivity
  - WLAN 802.11ac
  - USB 3.1 Gen 1
  - Bluetooth 5.0
- Storage
  - eMMC
  - Dual rank2 channel LPDDR4x memory support
- Multimedia
  - YUV + Bayer capture
  - sHDR v3.8
  - sHDR tuning through Chromatix
  - LDC
  - TNR (MCTF)

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4.2. New features

This section describes the new key software features that will be included in this SDK version.

- Up to 8 GB memory map (QCS610)
- Support WPA3-R3 certification on Helium

4.3. On-going features

- OTA Android 10 to 12
- Hastings chips and Android S for IOT SPs
- Boot change to improve cold boot latency for IoT
- Bring up LT6911UXC HDMI-IN

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Chapter 5. Test Report

Table 5-1. Sanity test report

Test case	Status
Fastboot flash	Pass
QFIL Flat flash	Pass
QFIL Meta flash	Pass
Recovery mode	Pass
Power on button	Pass
Volume up/down button	Pass
Force USB Boot button	Pass
Fastboot (adb & volume down)	Pass
BT/Wi-Fi/59 LED	Pass
LSM6DSOTR sensor	Pass
SD Card auto mount/unmount/multiple plug & unplug	Pass
Debug UART	Pass
Type A connection	Pass
Interaction between Type A and Type C	Pass
Type C connection	Pass
DP	Pass
LCD touch panel	Pass
MIPI LCD	Pass
Audio headphone	Pass
Audio speaker	Pass
BT connection/speed	Pass
Wi-Fi	Pass
Battery	NA
POE15W power supply	NA
12V DC power supply	Pass
Charge the battery via type-c	NA
Ethernet_Network port indicator	Pass
DHCP client	Pass
Sleep mode wakeup	Pass
System bootup time	Pass
Random read/write	Pass
IMX577 module capture & video	Pass
IMX415 module capture & video	Pass
IMX334 module capture & video	Pass

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Chapter 6. Known Issues

Table 6-1. List of known issues

Issue No.	Description	Status
Limitations-001	Not compatible with 2GB DDR	Open
Limitations-002	RJ45 indicator is abnormal	Open
Limitations-003	IMX577 module camera preview scale must 1:1	Open
BUG-9837	Portable U-disk not detected	Ongoing
BUG-8864	IMX577 module front camera cannot capture video	Ongoing
BUG-9835	The new display panel is unstable	Ongoing

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Appendix 1. Notices

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